Triple Integrals over a Circular Cylinder

Stewart, *Calculus: Early Transcendentals*, §15.7, Exercise 30

Problem

Express the integral

$$\iiint_E f(x, y, z)\, dV$$

as an iterated integral in six different ways, where E is the solid bounded by the surfaces

$$y^2 + z^2 = 9, \qquad x = -2, \qquad x = 2.$$

Part (a)

What the symbols mean

- E is the solid region of space. The integral adds up the values of f over all points of E.
- dV is the volume element. In Cartesian coordinates it is the product of the three differentials, for example $dV = dz\, dy\, dx$. The order of the differentials must match the order of the integral signs: the innermost sign goes with the first differential.
- f is a given function of three variables. In the answers below, f always means $f(x, y, z)$.
- An *iterated integral* is a chain of three ordinary integrals. The outer integral runs over a fixed interval. The middle integral runs over an interval that may depend on the outer variable. The inner integral runs between two surfaces, written as functions of the two variables that are already fixed.
- The *shadow* of E on a coordinate plane is the set of points (u, v) such that the line through (u, v) parallel to the remaining axis meets E. The shadow gives the limits of the two outer integrals.

Description of the solid

The equation $y^2 + z^2 = 9$ is a circle of radius 3 in the yz-plane. The same equation, with x free, is a circular cylinder of radius 3 whose axis is the x-axis. The planes $x = -2$ and $x = 2$ cut two circular caps from this cylinder. Therefore E is a right circular cylinder of radius 3 and length 4. In set notation,

$$\begin{aligned} E &= \{(x, y, z) : -2 \le x \le 2,\ y^2 + z^2 \le 9\} \\ &= \{(x, y, z) : -2 \le x \le 2,\ -3 \le y \le 3,\ -\sqrt{9 - y^2} \le z \le \sqrt{9 - y^2}\}. \end{aligned}$$

The second form solves the cylinder equation for z and shows the range of y.

For each fixed value $x = x_0$ with $-2 \le x_0 \le 2$, the section $E \cap \{x = x_0\}$ is the same disc

$$D = \{(y, z) : y^2 + z^2 \le 9\}$$

of area 9π. Figure 1 shows the solid, and Figure 2 shows the disc and one chord of it.

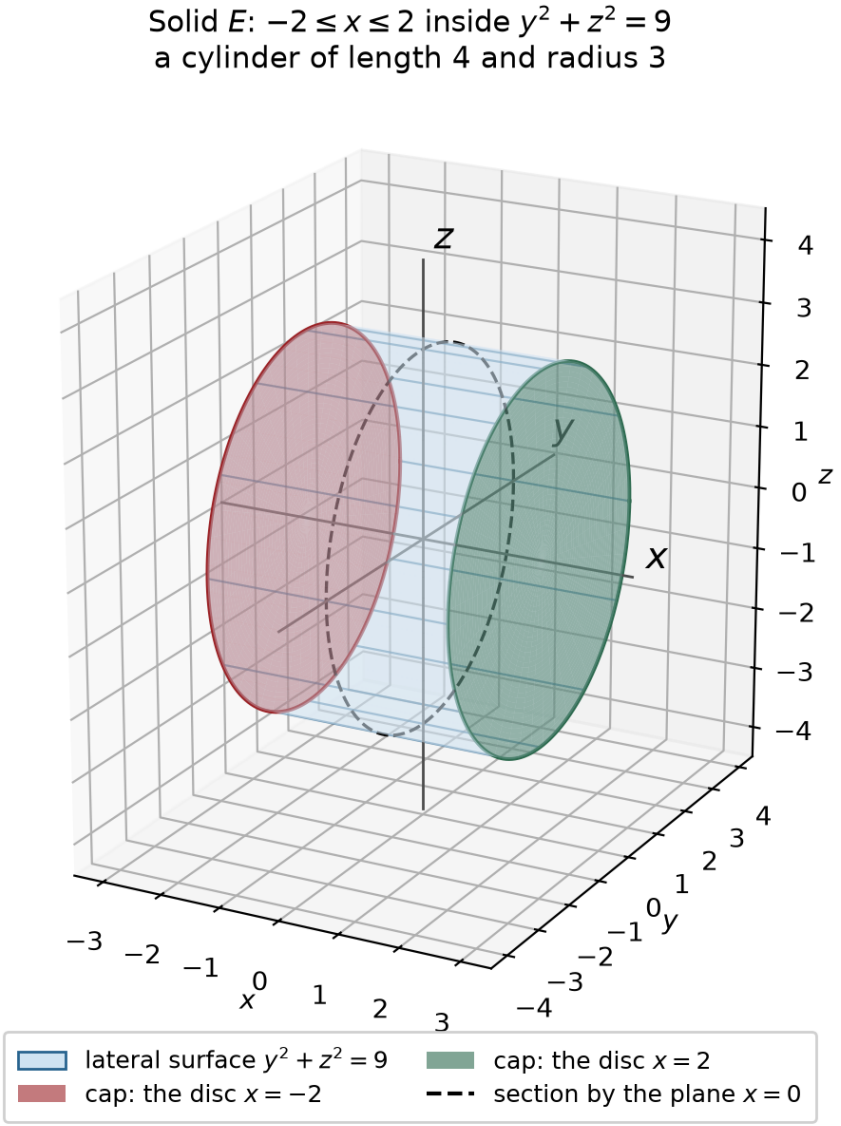


Figure 1: The solid E. The translucent tube is the cylinder $y^2 + z^2 = 9$. The red disc is the cap in the plane $x = -2$, and the green disc is the cap in the plane $x = 2$. The dashed circle is the section by the plane $x = 0$. The solid is a cylinder of radius 3 and length 4.

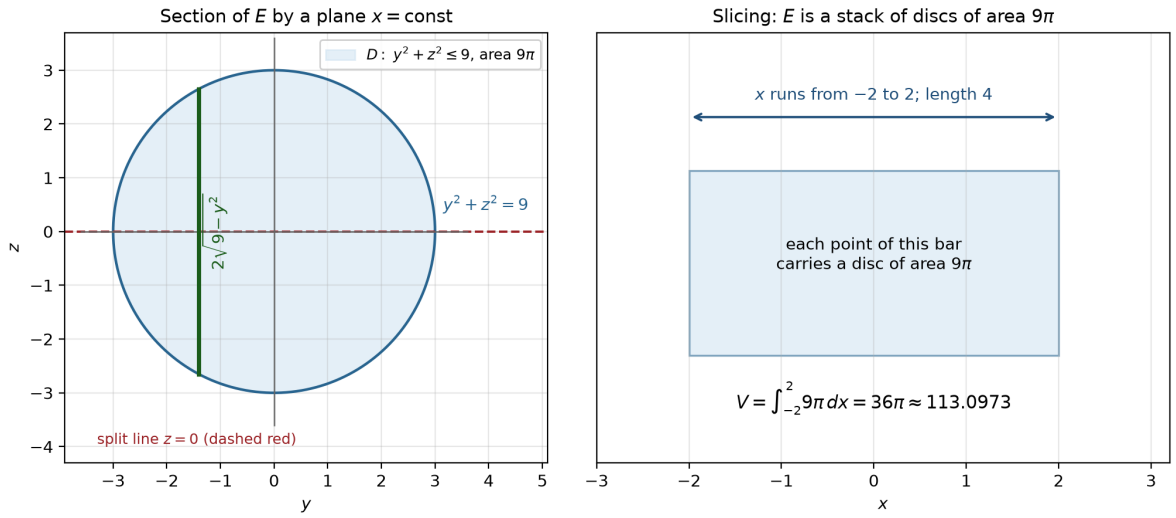


Figure 2: Left: the section of E by a plane $x = \text{const}$ is the disc $y^2 + z^2 \le 9$ of area 9π. The green chord is the piece of the vertical line $y = \text{const}$ that lies inside the disc; its length is $2\sqrt{9-y^2}$. The dashed red line $z = 0$ marks the two halves used by the two-piece form in the text. Right: the solid is a stack of such discs along the x-axis, so its volume is 9π times the length 4.

Volume as a check

Because every section has the same area 9π,

$$V = \iiint_E 1\, dV = \int_{-2}^{2} 9\pi\, dx = 9\pi \cdot 4 = 36\pi \approx 113.0973.$$

This number is used later as an independent check of all six orders.

The six iterated integrals

The rule for each order is the same. Pick the inner differential first. The two outer variables then run over the shadow of E on their coordinate plane. The inner limits come from the cylinder equation, solved for the inner variable.

(a) Order $dz\,dy\,dx$. The outer variable is x, so x runs from -2 to 2. The shadow on the xy-plane is the rectangle $-2 \le x \le 2$, $-3 \le y \le 3$. For fixed (x, y) the vertical line meets the cylinder where $z^2 = 9 - y^2$, that is at $z = \pm\sqrt{9-y^2}$. This is legal because the line meets E in exactly one segment.

$$\iiint_E f\, dV = \int_{-2}^{2} \int_{-3}^{3} \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f(x, y, z)\, dz\, dy\, dx.$$

(b) Order $dz\,dx\,dy$. The outer variable is now y, so y runs from -3 to 3. For each such y the section of E by the plane $y = \text{const}$ is the rectangle $-2 \le x \le 2$, $-\sqrt{9-y^2} \le z \le \sqrt{9-y^2}$. Hence

$$\iiint_E f\, dV = \int_{-3}^{3} \int_{-2}^{2} \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f(x, y, z)\, dz\, dx\, dy.$$

The inner limits are the same as in (a). Only the two outer integrals changed their order, and both of them have constant limits, so this step is valid.

(c) Order $dy\,dz\,dx$. Here y is the inner variable. Solve $y^2 + z^2 = 9$ for y and get $y = \pm\sqrt{9 - z^2}$. The shadow on the xz-plane is the rectangle $-2 \le x \le 2$, $-3 \le z \le 3$.

$$\iiint_E f\,dV = \int_{-2}^{2}\int_{-3}^{3}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f(x,y,z)\,dy\,dz\,dx.$$

(d) Order $dy\,dx\,dz$. This is the same as (c), with the two outer integrals exchanged. Both of them have constant limits.

$$\iiint_E f\,dV = \int_{-3}^{3}\int_{-2}^{2}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f(x,y,z)\,dy\,dx\,dz.$$

(e) Order $dx\,dy\,dz$. Now x is the inner variable, and its limits are the two planes: x runs from -2 to 2. The two outer variables run over the shadow of E on the yz-plane, and that shadow is the disc D, not the square $[-3,3] \times [-3,3]$.

$$\iiint_E f\,dV = \int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}\int_{-2}^{2} f(x,y,z)\,dx\,dz\,dy.$$

A common error here is to write $-3 \le z \le 3$ for the middle integral. That would add the four points of the square outside the disc, which are not in E.

(f) Order $dx\,dz\,dy$. This is the same as (e) with the two outer integrals exchanged, and the disc described from the z side.

$$\iiint_E f\,dV = \int_{-3}^{3}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}}\int_{-2}^{2} f(x,y,z)\,dx\,dy\,dz.$$

One order evaluated in full

Take order (a) and set $f = 1$. The inner integral is

$$\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} dz = 2\sqrt{9 - y^2}.$$

The middle integral is the area of the disc,

$$\int_{-3}^{3} 2\sqrt{9 - y^2}\,dy = 2 \cdot \frac{\pi \cdot 3^2}{2} = 9\pi,$$

because $\int_{-3}^{3} \sqrt{9 - y^2}\,dy$ is the area of the upper half of the disc. The outer integral gives

$$\int_{-2}^{2} 9\pi\,dx = 36\pi.$$

The value agrees with the slicing argument above.

A two-piece description of the same solid

The middle range of order (c) can be cut at $z = 0$. Then

$$\iiint_E f\,dV = \int_{-2}^{2}\int_{-3}^{0}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f\,dy\,dz\,dx + \int_{-2}^{2}\int_{0}^{3}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f\,dy\,dz\,dx.$$

Each piece is half of the cylinder, so each one has volume 18π, and the sum is 36π.

For this solid no order *needs* such a split. In all six orders the shadow is one region with a single description: a rectangle, or the disc. The inner variable is always trapped between two single graphs, $z = \pm\sqrt{9-y^2}$ or $y = \pm\sqrt{9-z^2}$, or between the two planes $x = \pm 2$. A split becomes necessary only when the shadow must be described by different formulas on different parts, or when the upper or the lower surface changes from one formula to another. Then the integral is cut into one piece per formula, and the pieces are added.

Verification

Three scripts check the answer. The file `part1_step1.py` integrates all six orders exactly with `sympy`, for $f = 1$ and for $f = x^2 + y^2 + z^2$, and also integrates the two-piece form of order (c). The file `part1_step2.py` computes the same six integrals by composite Gauss–Legendre quadrature with geometric grading near the circle, and compares the numbers with 36π and 210π. The file `part1_step3.py` tests 1,500,000 random points against the geometric inequalities and against the limits of each order, and finds no mismatch.

The second test function is $f = x^2 + y^2 + z^2$. Its exact value is

$$\iiint_E (x^2 + y^2 + z^2)\,dV = 48\pi + 81\pi + 81\pi = 210\pi \approx 659.7345.$$

The three terms come from the three coordinates: $\int_{-2}^{2} x^2\,dx \cdot 9\pi = 48\pi$, and $\int_D y^2\,dA \cdot 4 = 81\pi$, and the same for z^2.

Order of integration	$f = 1$	$f = x^2 + y^2 + z^2$
$dz\,dy\,dx$	113.097335529233	659.734457253857
$dz\,dx\,dy$	113.097335529233	659.734457253857
$dy\,dz\,dx$	113.097335529233	659.734457253857
$dy\,dx\,dz$	113.097335529233	659.734457253857
$dx\,dy\,dz$	113.097335529233	659.734457253857
$dx\,dz\,dy$	113.097335529233	659.734457253857
exact value	$36\pi = 113.097335529233$	$210\pi = 659.734457253857$

Table 1: Numerical values of the six iterated integrals (Gauss–Legendre quadrature, 40 graded panels with 12 nodes each). The spread over the six orders is 2.8×10^{-14} for $f = 1$ and 1.1×10^{-13} for the second function.

A Monte Carlo run with 6,000,000 samples in the box $[-2, 2] \times [-3, 3] \times [-3, 3]$ gives $V = 113.0983 \pm 0.0241$ and $Q = 659.607 \pm 0.205$. Both intervals contain the exact values, so the geometry of E is also confirmed by sampling.

Answer

$$\text{(a)}\quad \int_{-2}^{2}\int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f\,dz\,dy\,dx$$

$$\text{(b)}\quad \int_{-3}^{3}\int_{-2}^{2}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f\,dz\,dx\,dy$$

$$\text{(c)}\quad \int_{-2}^{2}\int_{-3}^{3}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f\,dy\,dz\,dx$$

$$\text{(d)}\quad \int_{-3}^{3}\int_{-2}^{2}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} f\,dy\,dx\,dz$$

$$\text{(e)}\quad \int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}\int_{-2}^{2} f\,dx\,dz\,dy$$

$$\text{(f)}\quad \int_{-3}^{3}\int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}}\int_{-2}^{2} f\,dx\,dy\,dz$$

Here f stands for $f(x,y,z)$. The volume of the solid is $\boxed{V = 36\pi}$.

How to solve problems of this kind

1. Draw the solid first. Name every bounding surface and find the curve where two of them meet. For this exercise the circle $y^2 + z^2 = 9$ is the meeting curve of the cylinder with each cap plane.
2. Find the shadow. The shadow on a coordinate plane gives the limits of the two outer integrals. A wrong shadow is the most common source of error. Here the shadow on the yz-plane is a disc, not a square.
3. Put the inner differential on the variable that is easiest to isolate. The inner limits are the two bounding surfaces written as functions of the two outer variables. For a cylinder, solving for z or for y is equally simple, so all six orders are available.
4. Test the inner limits. Move along a line parallel to the inner axis and check that it enters the solid once and leaves it once. If it enters or leaves more than once, split the region into pieces.
5. Check the answer with $f = 1$. The value must equal the volume, and the volume usually has a simple geometric formula. Here $V = \pi r^2 h = 9\pi \cdot 4$.
6. Check the answer with one non-constant f that has a known integral. A sum of squares is a good choice, because each coordinate can be integrated separately.
7. Confirm the six orders with a second method, for example numerical quadrature. The six numbers must agree with each other and with the exact value.
8. Keep the notation of the limits consistent. A limit may use only variables that are already fixed by an outer integral. In order $dz\,dy\,dx$ the inner limits may use y and x, but never z.